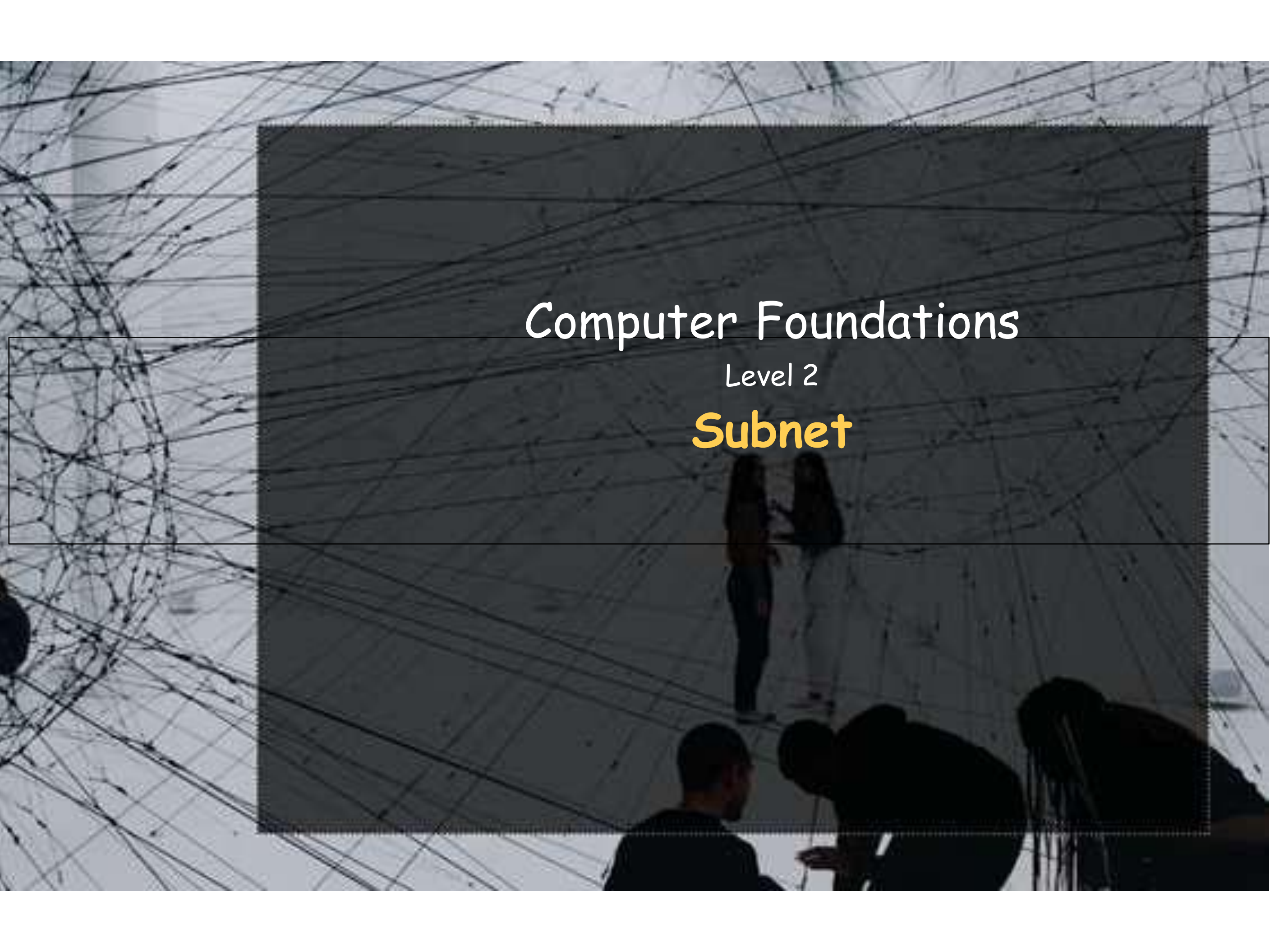




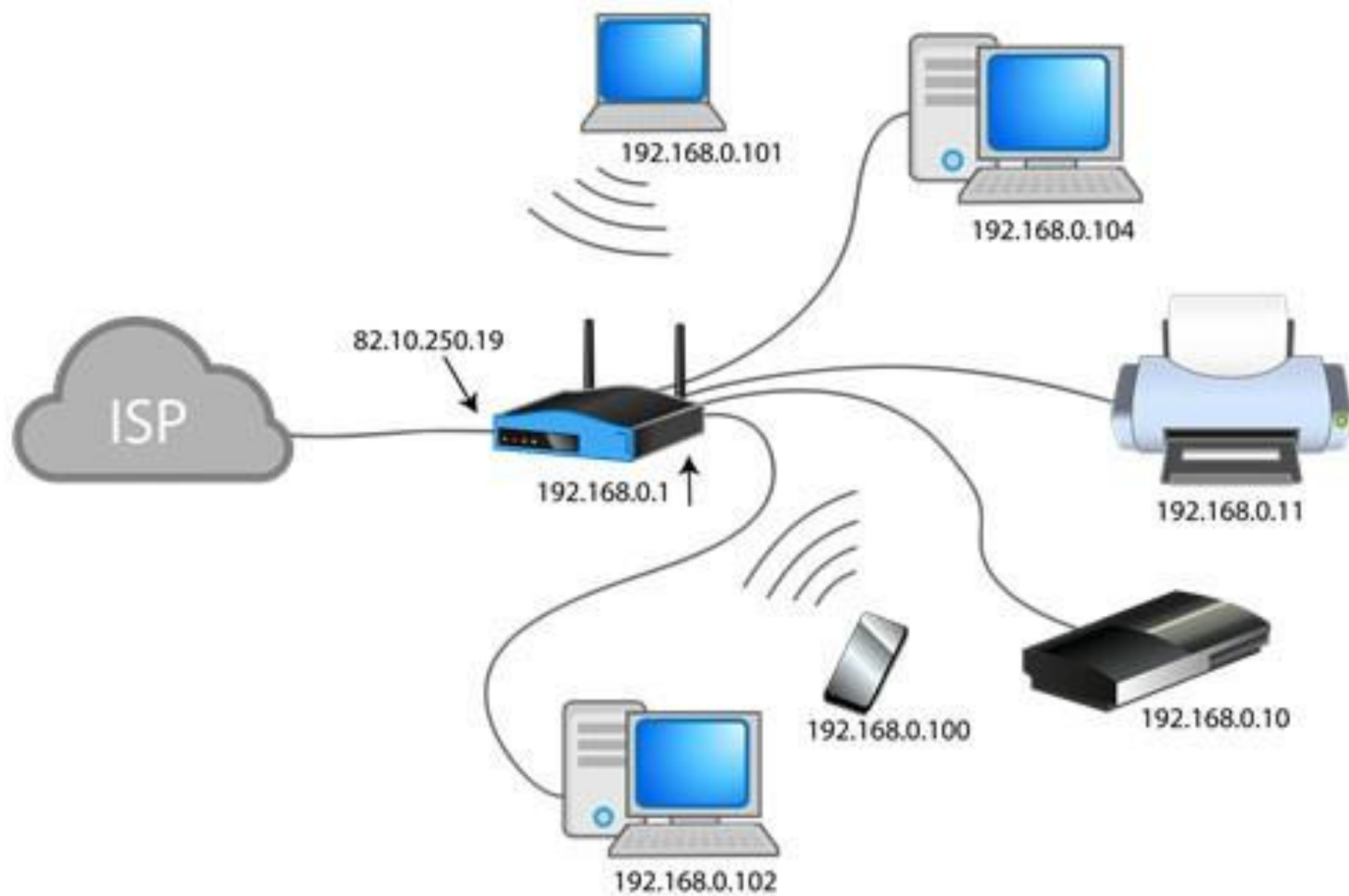
Computer Foundations Level 2

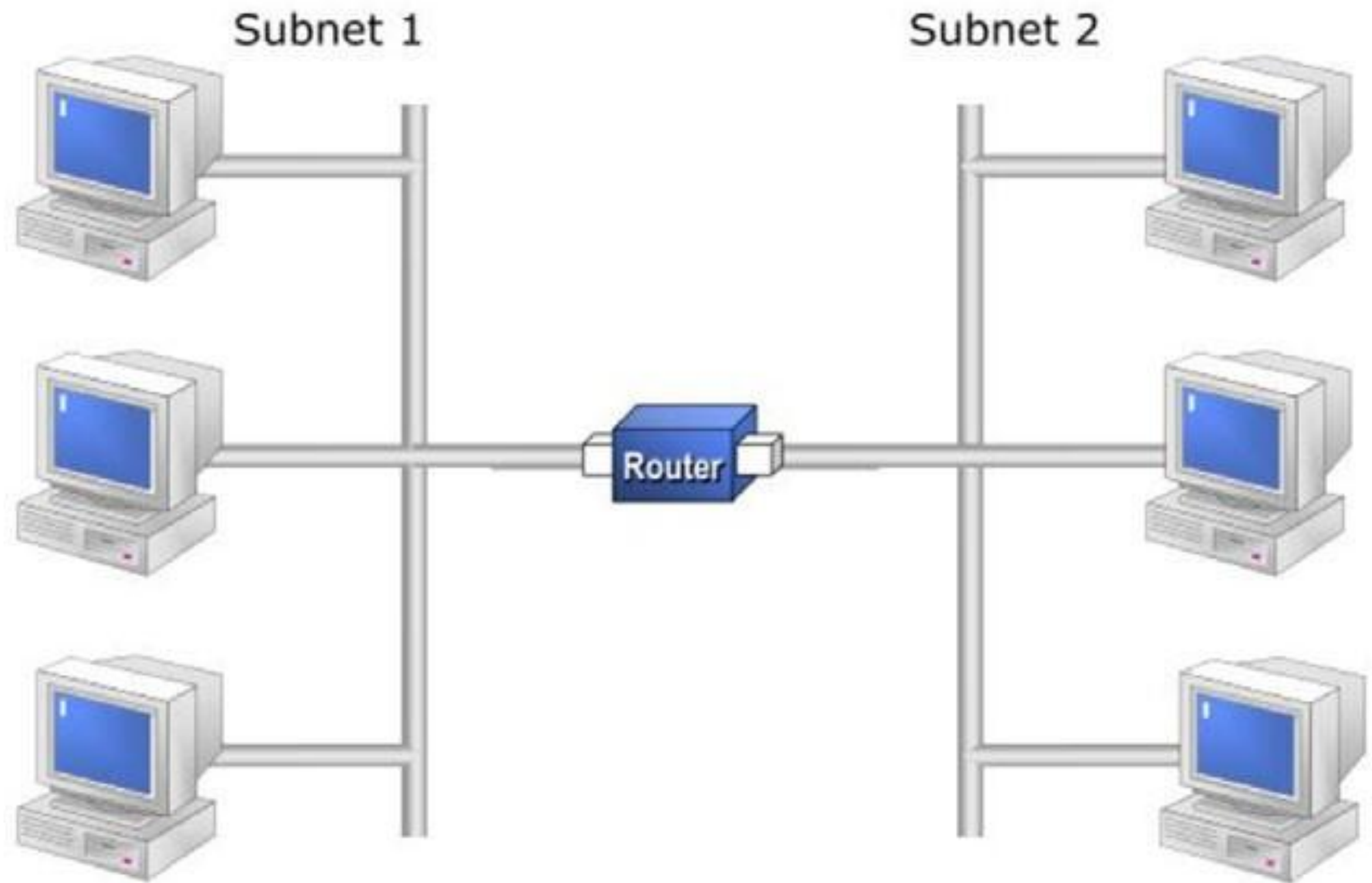


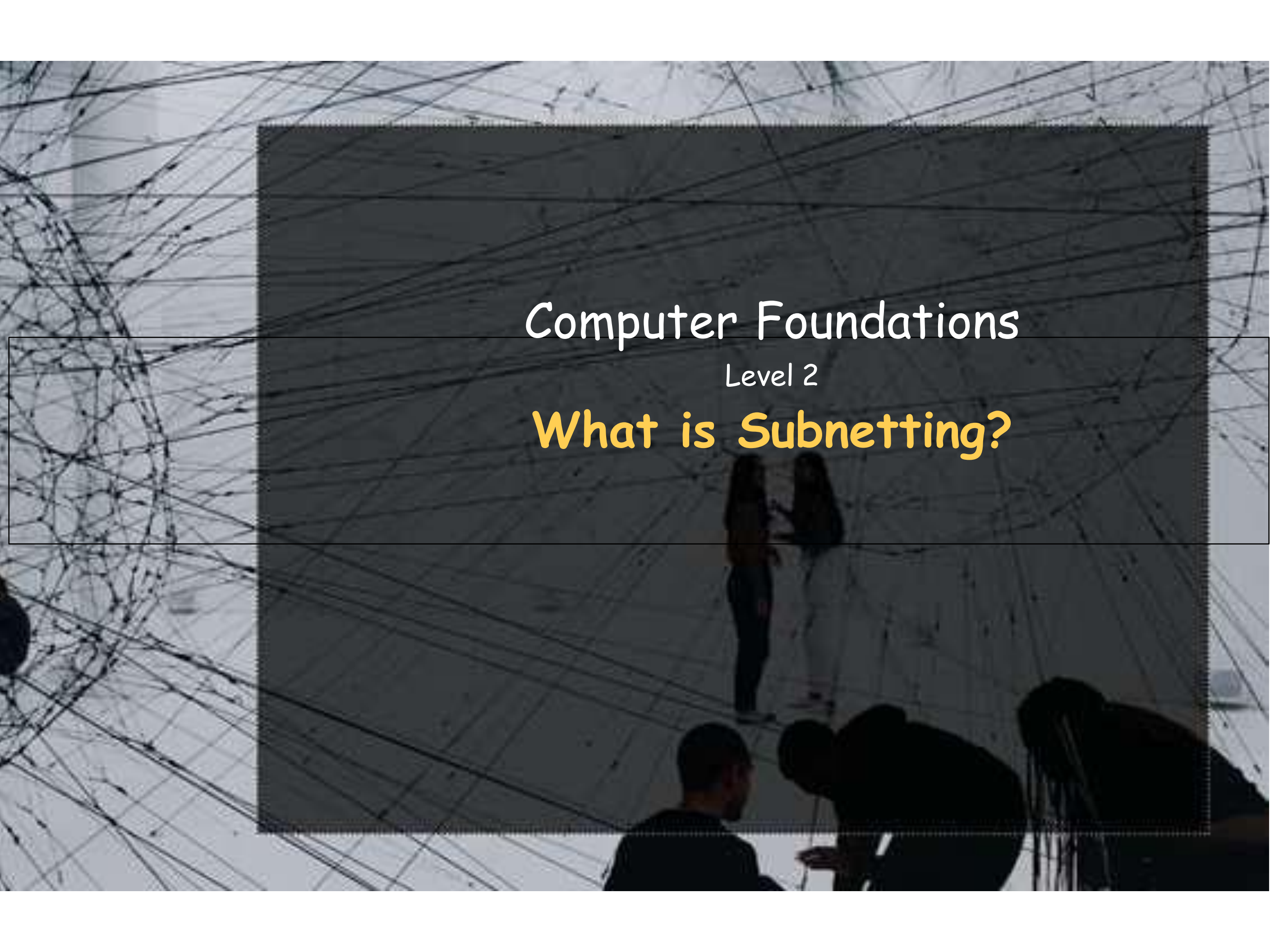
Computer Foundations

Level 2

Subnet





The background of the slide features a complex, abstract network of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a classroom setting. A large, dark, semi-transparent rectangular box is centered on the slide, serving as a backdrop for the text.

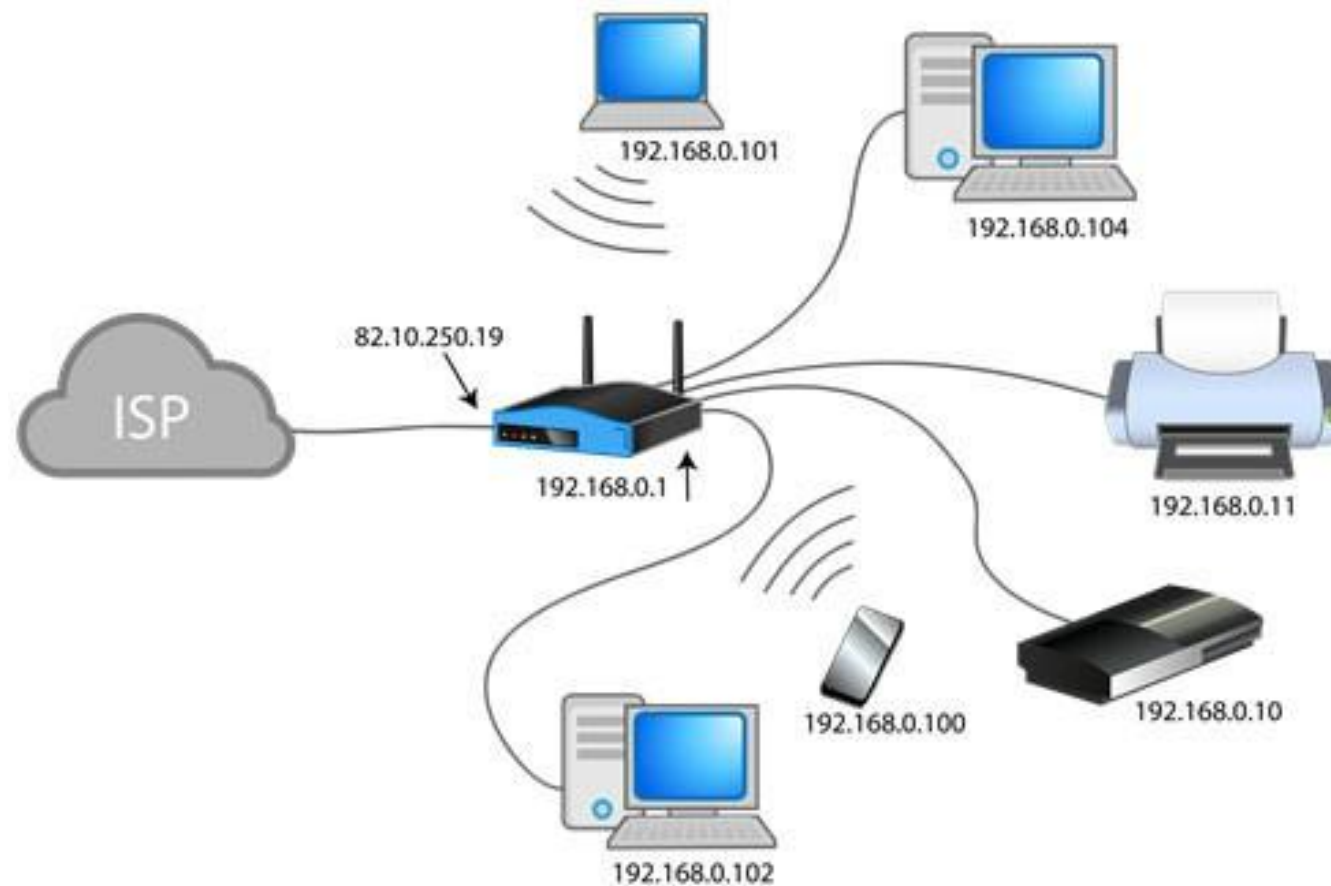
Computer Foundations

Level 2

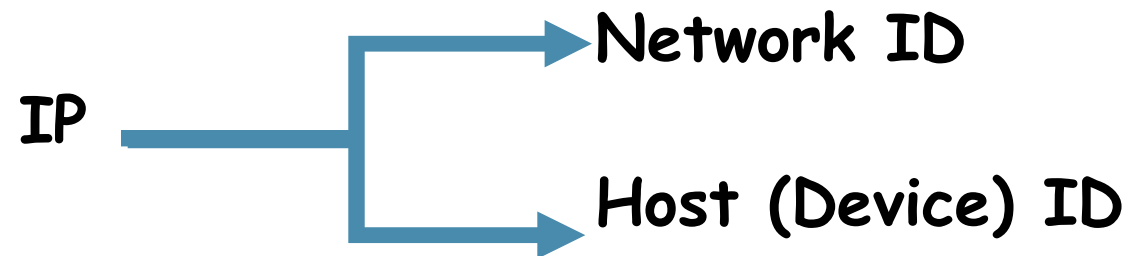
What is Subnetting?

What is Subnetting?

Subnetting is the practice of dividing a network into two or more smaller networks. It increases routing efficiency, enhances the security of the network.



203.177.100.2



Network Identifier: The first part of the IP address is the network identifier that identifies the network segment in which the computer is located. The same network identifier is required for all computers in the same segment, just as all houses in a particular county must have the same postal code.

Host identifier: The second part of the IP address is the host identifier, which identifies a computer, router, or other device within a segment. The host identifier must be unique within the network segment, just as the street name for a house within the postcode area must be unique.

Subnet Mask

The image shows a screenshot of the 'Internet Protocol (TCP/IP) Properties' dialog box, specifically the 'General' tab. The dialog box has a title bar with a question mark and a close button. Inside, there is a text box explaining that IP settings can be assigned automatically or manually. Two radio buttons are present: 'Obtain an IP address automatically' (unselected) and 'Use the following IP address:' (selected). Below the selected radio button, there are three input fields: 'IP address:' with the value '131 . 107 . 1 . 1', 'Subnet mask:' with the value '255 . 255 . 0 . 0', and 'Default gateway:' with the value '131 . 107 . 1 . 254'. Below these, there are two more radio buttons: 'Obtain DNS server address automatically' (unselected) and 'Use the following DNS server addresses:' (selected). Below the selected radio button, there are two input fields: 'Preferred DNS server:' with the value '131 . 107 . 1 . 66' and 'Alternate DNS server:' with the value ' . . .'. At the bottom right of the dialog box, there is an 'Advanced...' button. At the very bottom, there are 'OK' and 'Cancel' buttons.

Internet Protocol (TCP/IP) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 131 . 107 . 1 . 1

Subnet mask: 255 . 255 . 0 . 0

Default gateway: 131 . 107 . 1 . 254

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: 131 . 107 . 1 . 66

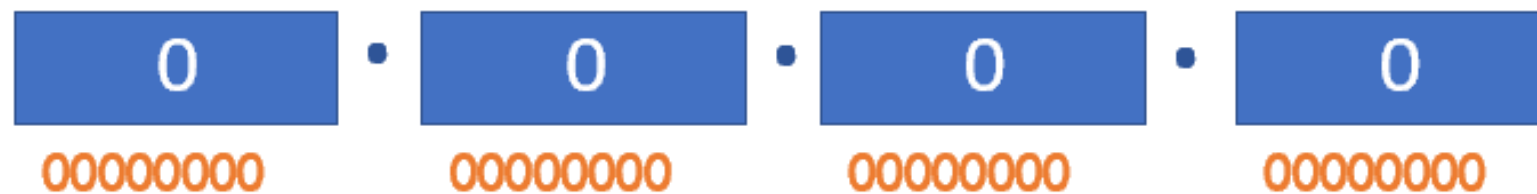
Alternate DNS server: . . .

Advanced...

OK Cancel

IP

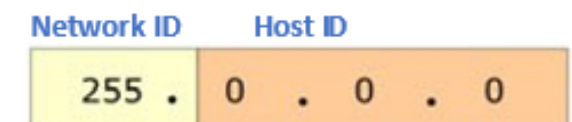
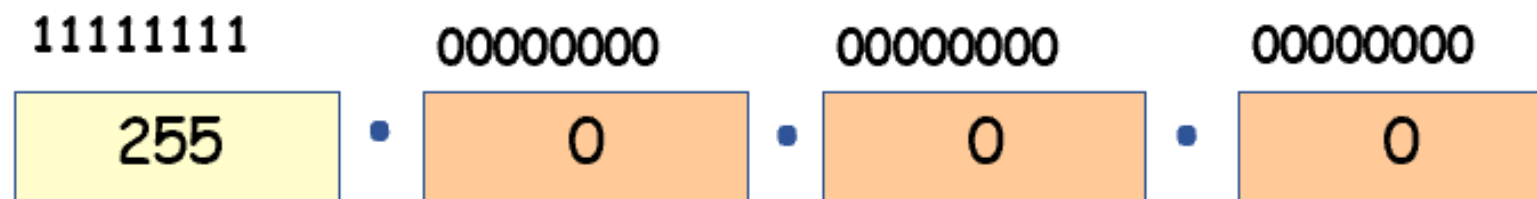
4 Bytes



CIDR (Classless Inter-Domain Routing)

Class A/8

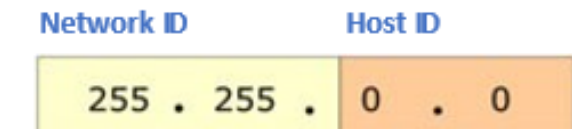
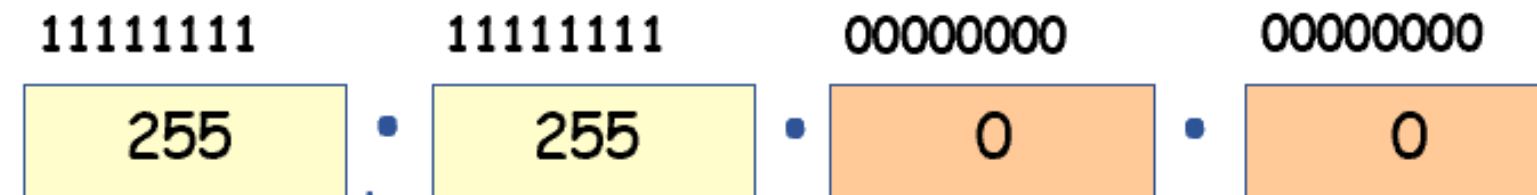
Subnet Mask



**2²⁴ IP Addresses
(16,777,216) Host**

Class B/16

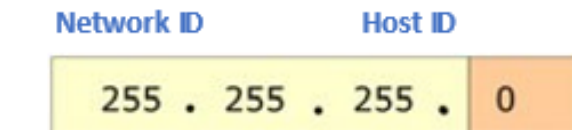
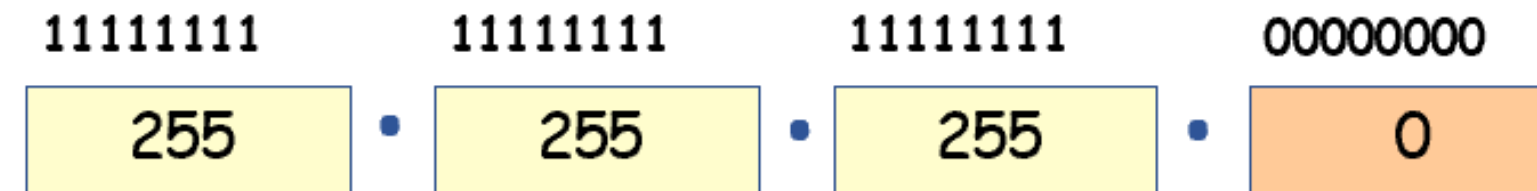
Subnet Mask



**2¹⁶ IP Addresses
(65,536) Host**

Class C/24

Subnet Mask



**2⁸ IP Addresses
(255) Host**

IP address:	131 . 107 . 1 . 1
Subnet mask:	255 . 255 . 0 . 0
Default gateway:	131 . 107 . 1 . 254

Subnet Mask

The image shows a screenshot of the 'Internet Protocol (TCP/IP) Properties' dialog box, specifically the 'General' tab. The dialog box has a title bar with a question mark and a close button. Inside, there is a 'General' tab selected. A text box explains that IP settings can be assigned automatically if the network supports it, or manually by asking a network administrator. There are two radio buttons: 'Obtain an IP address automatically' (unselected) and 'Use the following IP address:' (selected). Below the selected radio button, there are three input fields: 'IP address:' with the value '131 . 107 . 1 . 1', 'Subnet mask:' with the value '255 . 255 . 0 . 0', and 'Default gateway:' with the value '131 . 107 . 1 . 254'. Below these, there are two more radio buttons: 'Obtain DNS server address automatically' (unselected) and 'Use the following DNS server addresses:' (selected). Below the selected radio button, there are two input fields: 'Preferred DNS server:' with the value '131 . 107 . 1 . 66' and 'Alternate DNS server:' with the value ' . . .'. At the bottom right of the dialog box, there is an 'Advanced...' button. At the very bottom, there are 'OK' and 'Cancel' buttons.

Internet Protocol (TCP/IP) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 131 . 107 . 1 . 1

Subnet mask: 255 . 255 . 0 . 0

Default gateway: 131 . 107 . 1 . 254

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: 131 . 107 . 1 . 66

Alternate DNS server: . . .

Advanced...

OK Cancel

Thank you 😊